LAB 2: SYSTEM INFORMATION AND NETWORK DIAGNOSTICS

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1. Aim

To inspect the operating system, memory, disk space, network interfaces, routes, and connectivity using command-line tools.

2. Procedure

1. Opened Windows Command Prompt (CMD).
2. Executed commands to check the operating system version and computer name.
3. Viewed system details and memory information.
4. Listed the contents of the C: drive.
5. Inspected network interfaces and routing information.
6. Tested local TCP/IP connectivity using the loopback address.
7. Checked DNS resolution for "example.com".
8. Tested HTTPS connectivity using "curl".
9. Observed the outputs and recorded the findings.
10. Captured screenshots of the commands and their outputs as practical evidence.

3. Commands Used

S. No.| Task| Windows Command Prompt Command
1| OS version| "ver"
2| Host name| "hostname"
3| System details| "systeminfo"
4| Memory| "systeminfo"
5| Disk contents| "dir C:\"
6| IP interfaces| "ipconfig"
7| Routes| "route print"
8| Local TCP/IP| "ping -n 4 127.0.0.1"
9| DNS lookup| "nslookup example.com"
10| HTTPS connection| "curl -I --max-time 15 https://example.com"

4. Observations

1. Operating system: The Windows version was displayed using the "ver" command.

2. Total / available memory: The total and available physical memory values were displayed using "systeminfo".

3. Disk space: The "dir C:\" command displayed the files and directories present on the C: drive.

4. Active network interface: Network adapters and their configuration were displayed using "ipconfig".

5. Default gateway: The routing table was examined using "route print" to check for a default route.

6. Loopback replies: The local TCP/IP stack was tested using "ping -n 4 127.0.0.1". The actual number of successful replies was observed.

7. DNS resolution: The "nslookup example.com" command was used to check whether the domain name resolved to IP addresses.

8. HTTPS connection: The "curl" command was used to request HTTPS response headers. The displayed HTTP status or error was observed.

5. Interpretation

1. The "ver" command identifies the Windows version.
2. The "hostname" command displays the computer name.
3. The "systeminfo" command provides system configuration and memory information.
4. The "dir C:\" command lists the contents of the C: drive but does not reliably report available disk space.
5. The "ipconfig" command displays network interface configuration.
6. The "route print" command displays the routing table and helps identify the default gateway.
7. Successful loopback replies indicate that the local TCP/IP stack is responding; they do not prove Internet access.
8. DNS resolution and HTTPS connectivity are separate tests.
9. Ping may be blocked even when HTTPS works.

6. Result

The Windows command-line tools were executed to inspect system information, network configuration, routing, local TCP/IP connectivity, DNS resolution, and HTTPS connectivity.

7. Conclusion

This practical helped me understand how to use Windows Command Prompt for basic system inspection and network diagnostics. It also demonstrated the difference between local network testing, DNS resolution, and HTTPS connectivity.

8. Screenshots

Attach screenshots of the commands and their actual outputs below this heading.

Suggested screenshot labels:

1. Figure 1: Windows Version ("ver")
2. Figure 2: Computer Name ("hostname")
3. Figure 3: System Information ("systeminfo")
4. Figure 4: Disk Contents ("dir C:\")
5. Figure 5: Network Configuration ("ipconfig")
6. Figure 6: Routing Table ("route print")
7. Figure 7: Loopback Test ("ping -n 4 127.0.0.1")
8. Figure 8: DNS Lookup ("nslookup example.com")

```
C:\Users\Admin>ver

Microsoft Windows [Version 10.0.26200.9457]

C:\Users\Admin>hostname
LAPTOP-S3JR825J

C:\Users\Admin>systeminfo

Host Name:                     LAPTOP-S3JR825J
OS Name:                       Microsoft Windows 11 Home Single Language
OS Version:                    10.0.26200 N/A Build 26200
OS Manufacturer:               Microsoft Corporation
OS Configuration:              Standalone Workstation
OS Build Type:                 Multiprocessor Free
Registered Owner:              Admin
Registered Organization:       HP
Product ID:                    00342-42914-70552-AAOEM
Original Install Date:         11-08-2026, 00:07:30
System Boot Time:              29-09-2026, 23:13:41
System Manufacturer:           HP
System Model:                  Victus by HP Gaming Laptop 15-fb3xxx
System Type:                   x64-based PC
Processor(s):                  1 Processor(s) Installed.
                               [01]: AMD64 Family 25 Model 117 Stepping 2 AuthenticAMD ~3801 Mhz
BIOS Version:                  AMI F.15, 27-04-2026
Windows Directory:             C:\windows
System Directory:              C:\windows\system32
Boot Device:                   \Device\HarddiskVolume1
System Locale:                 en-us;English (United States)
Input Locale:                  00004009
Time Zone:                     (UTC+05:30) Chennai, Kolkata, Mumbai, New Delhi
Total Physical Memory:         23,691 MB
Available Physical Memory:     12,208 MB
Virtual Memory: Max Size:      25,227 MB
Virtual Memory: Available:     10,026 MB
Virtual Memory: In Use:        15,201 MB
Page File Location(s):         C:\pagefile.sys
Domain:                        WORKGROUP
Logon Server:                  \\LAPTOP-S3JR825J
```

9. Figure 9: HTTPS Connection ("curl -I --max-time 15 https://example.com")

```
                                    UEFI Code Readonly
                                    SMM Security Mitigations 1.0
                                    Mode Based Execution Control
                              Services Configured:
                                    Hypervisor enforced Code Integrity
                                    Secure Launch
                              Services Running:
                                    Hypervisor enforced Code Integrity
                              App Control for Business policy: Enforced
                              App Control for Business user mode policy: Enforced
                              Security Features Enabled:
Hyper-V Requirements:         A hypervisor has been detected. Features required for Hyper-V will not be displayed.

C:\Users\Admin>dir C:\
 Volume in drive C is Windows
 Volume Serial Number is 7EF4-5612

 Directory of C:\

16-09-2025  01:18    <DIR>          inetpub
01-04-2024  12:56    <DIR>          PerfLogs
10-10-2026  08:04    <DIR>          Program Files
30-09-2026  01:51    <DIR>          Program Files (x86)
10-08-2026  16:58    <DIR>          Users
29-09-2026  23:39    <DIR>          Windows
               0 File(s)              0 bytes
               6 Dir(s)  916,742,324,224 bytes free

C:\Users\Admin>ipconfig

Windows IP Configuration


Unknown adapter McAfee VPN:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi 2:

   Media State . . . . . . . . . . . : Media disconnected
```

```
   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi 3:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . : 2409:40d1:516:9f2:2335:b663:b6ad:b208
   Temporary IPv6 Address. . . . . . : 2409:40d1:516:9f2:bdbf:49ba:da14:2ed8
   Link-local IPv6 Address . . . . . : fe80::8a05:cb6f:4363:8135%17
   IPv4 Address. . . . . . . . . . . : 10.216.89.23
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::5431:f6ff:fe92:6379%17
                                       10.216.89.26

Ethernet adapter Bluetooth Network Connection:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

C:\Users\Admin>route print
===========================================================================
Interface List
  9...00 ff 5f 4b 22 ff ......TAP-Windows Adapter V9
 11...22 58 43 d1 cb c0 ......Realtek 8852BE-VT Wireless LAN WiFi 6 PCI-E NIC #2
 12...26 58 43 d1 cb c0 ......Realtek 8852BE-VT Wireless LAN WiFi 6 PCI-E NIC #3
```

```
 17     66 ::/0                     fe80::5431:f6ff:fe92:6379
  1    331 ::1/128                  On-link
 17     66 2409:40d1:516:9f2::/64   On-link
 17    306 2409:40d1:516:9f2:2335:b663:b6ad:b208/128
                                    On-link
 17    306 2409:40d1:516:9f2:bdbf:49ba:da14:2ed8/128
                                    On-link
 17    306 fe80::/64                On-link
 17    306 fe80::8a05:cb6f:4363:8135/128
                                    On-link
  1    331 ff00::/8                 On-link
 17    306 ff00::/8                 On-link
===========================================================================
Persistent Routes:
  None

C:\Users\Admin>ping -n 4 127.0.01

Pinging 127.0.0.1 with 32 bytes of data:
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128

Ping statistics for 127.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\Admin>nslookup example.com
Server:  UnKnown
Address:  10.216.89.26

Non-authoritative answer:
Name:    example.com
Addresses:  2606:4700:90c5:72db:f264:560:ef6b:ff98
          104.20.23.154
          172.66.147.243


C:\Users\Admin>curl -I --max-time 15 https://example.com
```